

Research note: a mechanism for the SMS–BCE gap in OIT

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Summary

Section 3.4 of *Organized Information Transmission* (Mathevet & Taneva, 2022) (hereafter OIT; authors hereafter M&T) reports that two BCE refinements in an organisational context, single-meeting scheme (SMS) and the delegated hierarchy (DH), attain approximately 95% of the less-restricted BCE optimum in a specific numerical example, and leaves open the mechanism behind this close performance. This note extends M&T’s Section 3.4 computation through a 2000-draw parameter sweep and proposes a mechanism conjecture for the gap when it exists. The conjecture: cross-agent heterogeneity in the cost/benefit ratio enlarges the SMS–BCE gap once the gap is active; what activates it is left open.

Computational results

I re-solved the LP problem for BCE and SMS across 2000 settings of parameters drawn from $\gamma_{11} \in [0.5, 1.5]$, $\gamma_{22} \in [0.33, 1.33]$, $\gamma_{33} \in [0.25, 1.25]$, $\gamma_{ij} \in [0, 0.15]$, and $c_i \in [0.5, 1.5]$ for each agent. The draws cover a wide range of cost-benefit ratios c_i/γ_{ii} across agents. The BCE–SMS gap lies below a 10^{-6} tolerance in 90.85% of draws; 183 draws have a non-trivial positive gap. The 95% relative efficiency reported in M&T’s Section 3.4 sits close to the least-performing case in this sweep; mean and median SMS efficiency in the positive-gap subsample are both above 98.5%.

Statistic	Value
Parameter draws (total)	2,000
Draws with positive gap (above 10^{-6} tolerance)	183
Frequency of gap below tolerance (of total)	0.9085
Worst-case SMS efficiency	0.9431
Mean SMS efficiency, positive-gap subsample	0.9862
Median SMS efficiency, positive-gap subsample	0.9896
Maximum relative gap, positive-gap subsample	0.0569
Maximum absolute gap (same draw)	0.0725

Table 1: Headline statistics, 2,000 random parameter draws.

Mechanism

M&T use the cost/benefit ratio (c_i/b_i in M&T’s regime-change notation, c_i/γ_{ii} denotes the counterpart in the Section 3.4 example) as a tool to identify the agent who should go to the lower part of

the delegated hierarchy in their regime-change application (Example 4): agents with lower c_i/b_i play the high action more often and occupy lower positions in the hierarchy. The conjecture here is that the same kind of heterogeneity drives the SMS–BCE gap, via the SMS joint obedience constraint.

Within the 183-draw non-trivial-gap subsample, the relative gap has a correlation of 0.552 with the cross-agent standard deviation of c_i/γ_{ii} , and -0.519 with the level of the BCE optimum. These correlations are conditional on the gap being positive; in the full sample they disappear, since the gap is zero (within tolerance) in 90.85% of draws, where SMS attains the BCE optimum exactly and the equal-informedness restriction is costless. The dispersion measure should therefore be read as governing the size of the loss when the SMS restriction is costly, not as a full-sample predictor of the gap.

Predictor (within positive-gap subsample, $n = 183$)	Correlation with rel. gap
SD of c_i/γ_{ii} across agents	0.552
Count of hard agents ($\gamma_{ii}/c_i \leq 1$)	0.367
BCE objective value (level)	-0.519

Table 2: Correlations with the relative gap, positive-gap subsample.

One interpretation: when there are agents with high benefit-cost ratio γ/c (easier to persuade) and with low benefit-cost ratio (harder to persuade), the designer has to reduce the weight on the situations in which the easy agent plays effort = 1 and the hard agent plays zero effort to maintain the obedience constraint when agents in the meeting observe the same message. This effect would not turn on if all agents are hard or easy (homogeneous agents).

Open issues

Three questions remain open. First, whether heterogeneity in c/γ also predicts *when* the gap is positive (the extensive margin), rather than only its size conditional on being positive; the current sweep does not settle this. Second, whether “heterogeneity in c/γ activates SMS joint obedience” is the right formalisation of the mechanism, or whether a sharper object (a different dispersion measure, or the binding constraint expressed in dual variables) drives the gap. Third, prior literature on heterogeneity effects in OIT-style refinements of BCE that I may be missing.